

JEE MAIN REVISION MASTER: SURFACE CHEMISTRY 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 19 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Surface Chemistry — Standard Calculations & Problem Archetypes	9 PYQs	Quantitative Analysis
High	Surface Chemistry — Applied Cases, Reactions & Traps	9 PYQs	Applied Topics
Moderate	Surface Chemistry — Core Principles & Governing Laws	3 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Surface Chemistry — Standard Calculations & Problem Archetypes	$BaCl_2, AlCl_3, \log \frac{x}{m} = \log k + \frac{1}{n} \log p, SO_2$
Surface Chemistry — Applied Cases, Reactions & Traps	$I^-, AgI/Ag^+, [Fe(CN)_6]^{4-} \rightarrow PO_4^{3-} \rightarrow SO_4^{2-} \rightarrow Cl^-$
Surface Chemistry — Core Principles & Governing Laws	$\Delta H, \Delta G = \Delta H - T \Delta S$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Surface Chemistry — Standard Calculations & Problem Archetypes (High Yield — 9 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Surface Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Shift] Q#1 (MCQ): "For Freundlich adsorption isotherm, a graph of $\log(x/m)$ vs $\log(p)$ gives a straight line. The slope of line and its Y-axis intercept $r...$ " [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "Aqueous solution of raw sugar when passed over beds of animal charcoal, it becomes colourless. Pick the correct set of terminologies that ca..." [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): "Zeta potential is..." [Ans: N/A]

3.2 Surface Chemistry — Applied Cases, Reactions & Traps (High Yield — 9 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Surface Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Identify the incorrect statement..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "For the coagulations of positively charged hydrated ferric - oxide sol, the flocculating power of the ions is in the order:..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "Colloidal solution commonly used in the treatment of skin disease is..." [Ans: N/A]

3.3 Surface Chemistry — Core Principles & Governing Laws (Moderate Yield — 3 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Surface Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Surface Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Surface Chemistry PYQs.

Representative Source PYQs:

- [2024 Shift] Q#3 (MCQ): "Gold sol is not a..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "During adsorption of a gas on a solid at low temperature..." [Ans: N/A]
- [2017 Shift] Q#1 (MCQ): "Which of the following is not a favourable condition for physical adsorption?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Identify the incorrect statement

- (A) Values of colligative properties of colloidal solution are of small order compared to values of true solution.
- (B) Tyndall effect is observed only when diameter of the dispersed particles is not much smaller than wavelength of incident light.
- (C) Colour of colloidal solution depends on the wavelength of light scattered by the dispersed particles.
- (D) Brownian movement is due to balanced bombardment of molecules of dispersion medium on colloidal particles.

Correct Answer: None

Solution Analysis: Among the given statements incorrect statement is given in option (d). Its correct form is "Brownian movement is due to unbalanced bombardment of molecules of dispersion medium on colloidal particles".

Benchmark Problem 2: KCET 2023 (Q#1)

For Freundlich adsorption isotherm, a graph of $\log(x/m)$ vs $\log(p)$ gives a straight line. The slope of line and its Y-axis intercept respectively are

- (A) $\log((1/n)^k)$
- (B) $(1/n), \log k$
- (C) $\log((1/n)), \log k$
- (D) $(1/n), k$

Correct Answer: None

Solution Analysis: We know that, Freundlich adsorption isotherm equation is given by $\log(x/m) = \log k + (1/n) \log p$ (i)
Also equation of straight line is $y = mx + c$ (ii) On comparing Eq. (i) and Eq. (ii), we get Slope, $m = (1/n)$ Intercept, $c = \log k$

Benchmark Problem 3: KCET 2020 (Q#3)

During adsorption of a gas on a solid at low temperature

- (A) $\Delta G < 0, \Delta H < 0, \Delta S < 0$
- (B) $\Delta G > 0, \Delta H > 0, \Delta S > 0$
- (C) $\Delta G < 0, \Delta H < 0, \Delta S > 0$
- (D) $\Delta G < 0, \Delta H > 0, \Delta S < 0$

Correct Answer: None

Solution Analysis: Adsorption is an exothermic process with $\Delta H =$ negative and since, gas is being adsorbed to a solid the randomness of the gas is decreasing making $\Delta S < 0$. Also, $\Delta G = \Delta H - T\Delta S$ Here, $T\Delta S$ is small negative value at low temperature and hence, $\Delta G < 0$. Which supports the spontaneity of the adsorption.